

ESB-R

ELECTRIFIED HELICAL CALCINER

HIGH-TEMPERATURE LABORATORY CALCINATION SYSTEM

*Advanced Electrified Helical Downtube
Technology for Rapid Sample Processing*

APPLICATIONS

-  Cement Research Laboratories
-  Mineral Processing
-  Lime Industry
-  Universities & Research Institutes
-  Material Science
-  Process Development
-  Pilot Plant Testing

KEY FEATURES

- ✓ Electrically Heated Helical Downtube Technology
- ✓ Rapid Sample Calcination
- ✓ Uniform High Temperature Heating
- ✓ Compact Laboratory Design
- ✓ Heavy Duty Insulated Construction
- ✓ Precise Temperature Control
- ✓ Designed for Continuous Research Applications

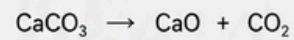


WORKING PRINCIPLE

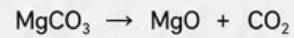
- 1  Material is manually fed into the top hopper.
- 2  Sample enters the stainless-steel helical downtube.
- 3  High-temperature electric heaters uniformly heat the material.
- 4  Calcination reaction occurs during downward movement.
- 5  CO₂ is released through the vent.
- 6  Calcined product is collected at the bottom discharge tray.

TYPICAL CALCINATION REACTIONS

LIMESTONE



MAGNESITE



MAJOR ADVANTAGES

- ✓ Uniform Heat Distribution
- ✓ Excellent Thermal Efficiency
- ✓ Fast Heating Rate
- ✓ Compact Footprint
- ✓ Low Heat Loss
- ✓ Easy Maintenance
- ✓ Modular Construction
- ✓ High Temperature Stability
- ✓ Industrial Grade Construction
- ✓ Long Service Life



High
Temperature
Operation



Rapid
Calcination



Helical
Downtube
Technology



Precise
Temperature
Control



Heavy Duty
Insulated
Construction



Low
Maintenance



Continuous
Research
Applications

ESB-R

ELECTRIFIED HELICAL CALCINER

PRODUCT OVERVIEW

The ESB-R Electrified Helical Calciner is a compact high-temperature laboratory calcination system designed for rapid thermal decomposition of limestone and other mineral samples.

The system utilizes a specially engineered helical downtube housed inside a heavily insulated electrically heated chamber, enabling uniform heating and excellent temperature stability.

The system is ideal for research organizations, cement industries, universities, pilot plants, and advanced material testing laboratories requiring repeatable and accurate calcination experiments.

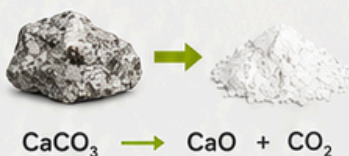


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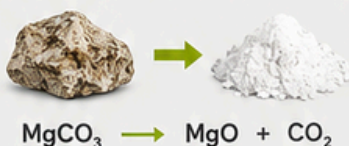
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PARAMETER	SPECIFICATION
System Type	Electrified Helical Downtube Calciner
Operation	Batch Type
Feed Capacity	10–100 g per batch
Material	CaCO ₃ , MgCO ₃ and similar minerals
Heating Method	Electric Resistance Heating
Heating Element	Kanthal A-1
Element Temperature	Up to 1500°C
Calcination Temperature	850–900°C (Application Dependent)
Reactor Material	SS304 / SS316
Helical Pipe OD	30 mm
Helical Pipe ID	25 mm
Helical Pipe Length	Approx. 4 m
Helical Pitch	Approx. 600 mm
Residence Time	3–4 Seconds
Heating Power	10–12 kW (Typical)
Temperature Sensors	6 × K-Type Thermocouples
Chamber Insulation	Dense Refractory + 50 mm Glass Wool
Chamber Size	600 × 600 mm
Overall Height	Approx. 3300 mm
Control System	PLC / PID Based Temperature Control (Optional)
Safety Features	Emergency Stop, Earthing, Over Temperature Protection



WHY ENERSOL BIOPOWER?

- ✓ In-house Design & Manufacturing
- ✓ High Temperature Process Expertise
- ✓ Customized Engineering Solutions
- ✓ Advanced Thermal System Design
- ✓ Strong R&D Team
- ✓ Installation & Commissioning Support
- ✓ After-Sales Service
- ✓ Customized Automation Options



OPTIONAL ACCESSORIES

- PLC with HMI
- SCADA Monitoring
- Data Logging Software
- Variable Temperature Profiles
- Industrial Safety Interlocks
- Customized Sample Handling



CONTACT US

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Where Waste Becomes Power

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